
CIS 678 - Machine Learning

— Probability and Maximum Likelihood Estimation (MLE) —

— Instructor: Yong Zhuang —

yong.zhuang@gvsu.edu

Last Class

- Why Learn from Data?
 - Traditional programming and machine learning
- Instance-based and model-based learning
- How Model-Based Learning Works
 - Model selection, training, and prediction
- Data challenges
- Generalization: Underfitting and overfitting
- Training/validation/test data
- Diagnosing Data Mismatch: **Training, Train-dev, Dev(validation), Test**
- Leakage-safe evaluation

Agenda

- Probabilistic View of Machine Learning
- Joint and Conditional Probability
- Probability vs. Likelihood
- Maximum Likelihood Estimation (MLE)
 - MLE for Bernoulli Variables
 - Sample Complexity and High-Probability Bounds

Joint Distribution

What does this has to do with function approximation?

Instead of learning $F:X \rightarrow Y$, learn $P(Y|X)$



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What does this has to do with function approximation?

Can design algorithms that learn functions with uncertain outcomes (e.g., predicting tomorrow's stock price) and that incorporate prior knowledge to guide learning (e.g., a bias that tomorrow's stock price is likely to be similar to today's price).

Joint Distribution

The key to building probabilistic models is to define a set of random variables, and to consider the joint probability distribution over them

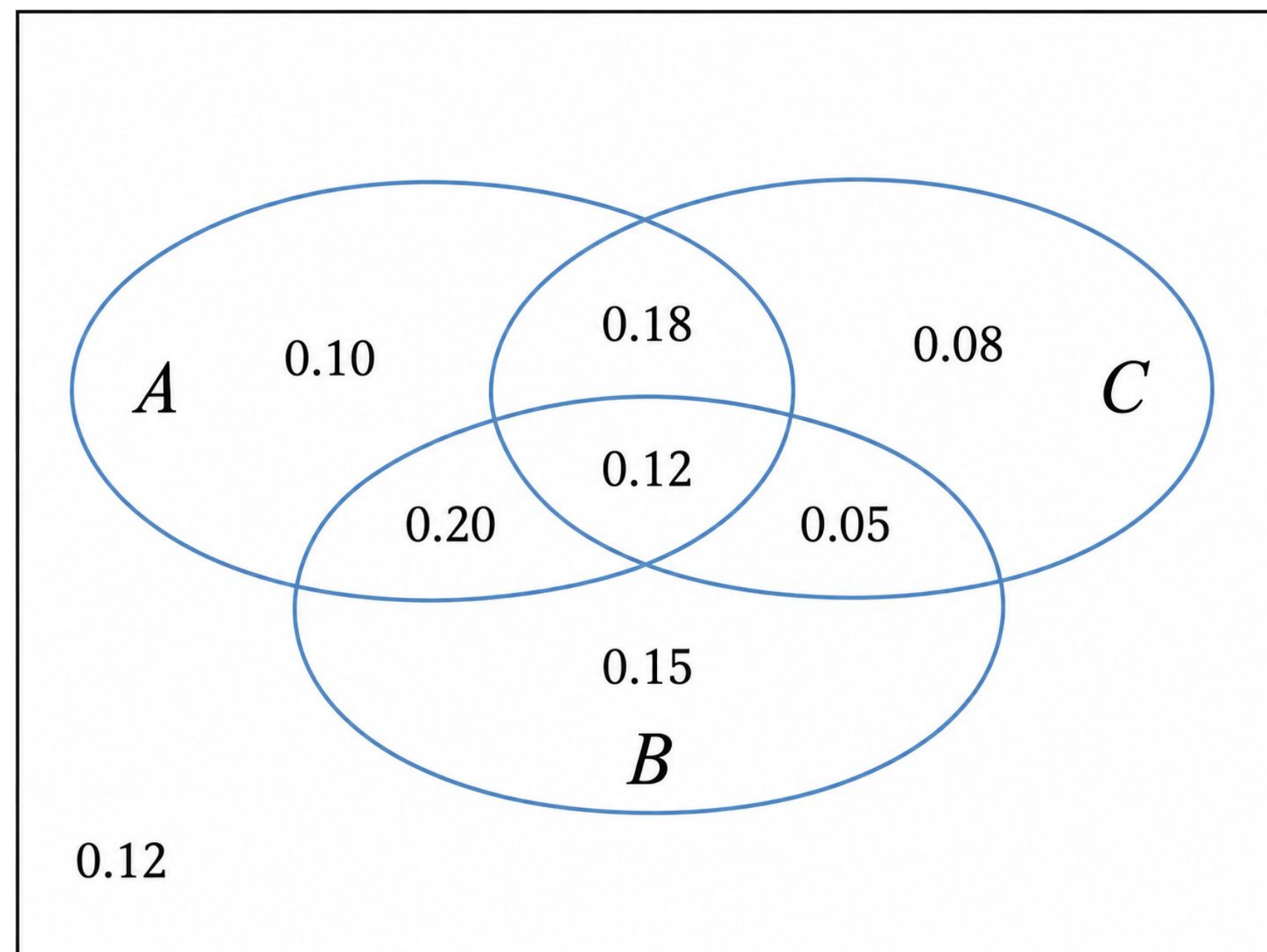
A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Joint Distribution

Recipe for making a joint distribution of M variables:

- Make a truth table listing all combinations of values (M Boolean variables $\rightarrow 2^M$ rows).
- For each combination of values, say how probable it is.
- By the axioms of probability, these probabilities must sum to 1.



A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
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1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Using the Joint Distribution

Once we have the Joint Distribution, can ask for the probability of any logical expression involving these variables

$$P(E) = \sum_{\text{rows matching } E} P(\text{row})$$

$$P(C) = ?$$

$$P(A) = 0.08 + 0.05 + 0.18 + 0.12 = 0.43$$

A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
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Using the Joint Distribution

Once we have the Joint Distribution, can ask for the probability of any logical expression involving these variables

$$P(E) = \sum_{\text{rows matching } E} P(\text{row})$$

$$P(A \ \& \ C) = ?$$



$$P(A \ \& \ C) = 0.18 + 0.12 = 0.3$$

A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
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1	1	1	0.12

Example: Boolean variables A,B,C

Inference with the Joint Distribution

Once we have the Joint Distribution, can ask for the probability of any logical expression involving these variables

$$P(E_1 \mid E_2) = \frac{P(E_1 \wedge E_2)}{P(E_2)} = \frac{\sum_{\text{rows matching } E_1 \text{ and } E_2} P(\text{row})}{\sum_{\text{rows matching } E_2} P(\text{row})}$$

$$P(A \mid C) = ?$$



$$P(A \mid C) = 0.3 / 0.43 = 0.698$$

A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Learning and the Joint Distribution

Suppose we want to learn the function

$$F:A, B \rightarrow C$$

||

$$P(C \mid A, B)$$

One solution learn joint distribution from data, calculate $P(C \mid A, B)$

e.g., $P(C=1 \mid A=1, B=1) = ?$



A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
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1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Learning and the Joint Distribution

Suppose we want to learn the function

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One solution learn joint distribution from data, calculate $P(C \mid A, B)$

e.g., $P(C=1 \mid A=1, B=1) = ?$

$$= 0.12 / (0.12 + 0.2) = 0.375$$



A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Idea: learn classifiers by learning $P(Y|X)$

Consider:

- $Y=C$,
- $X = (A, B)$

A	B	$P(C=1 A,B)$	$P(C=0 A,B)$
0	0	0.4	0.6
0	1	0.25	0.75
1	0	0.643	0.357
1	1	0.375	0.625

A	B	C	Prob
0	0	0	0.12
0	0	1	0.08
0	1	0	0.15
0	1	1	0.05
1	0	0	0.10
1	0	1	0.18
1	1	0	0.20
1	1	1	0.12

Example: Boolean variables A,B,C

Probability vs. Likelihood

Probability

Given a fixed parameter θ , ask:

How probable is a particular outcome or dataset?

$$P(D | \theta)$$

- θ is fixed
- D is variable
- Probability describes possible outcomes under a model

Likelihood

After observing data D , ask:

Which values of θ best explain the observed data?

$$L(\theta; D) = P(D | \theta)$$

- D is fixed
- θ is variable
- Larger likelihood means the observed data are more compatible with that parameter value

Probability vs. Likelihood

Probability vs. Likelihood: Coin Example

Suppose the observed sequence is

$$D = \{H, H, T\}$$

so

$$\alpha_H = 2, \quad \alpha_T = 1.$$

For a coin with parameter θ :

$$P(D \mid \theta) = \theta^2(1 - \theta)$$

Probability view:

If $\theta = 0.5$, what is the probability of observing H, H, T ?

$$P(D \mid \theta = 0.5) = (0.5)^2(0.5) = 0.125$$

Likelihood view:

We already observed H, H, T . Which θ makes this observation most likely?

$$L(\theta) = \theta^2(1 - \theta)$$

Estimating Probabilities from Data

Estimating the Bias of a Coin

Problem: Assume we can flip a coin with bias θ several times. Estimate the probability that it turns out heads when we flip it?

Each flip yields a Boolean value for X , $X \sim \text{Bernoulli}(\theta)$

Bernoulli Random Variable: $P(X=1) = \theta$; $P(X=0) = 1-\theta$



$X=1$

$X=0$

We flip it repeatedly, observing the outcome:

- It turns Heads (i.e. $X=1$) α_H times
- It turns Tails (i.e. $X=0$) α_T times

How can we estimate the probability of heads $\theta = P(X=1)$?



Maximum Likelihood Estimation(MLE)

Problem: Assume we can flip a coin with bias θ several times. Estimate the probability that it turns out heads when we flip it?

Maximum Likelihood Estimation

Choose parameter $\hat{\theta}$ that maximizes likelihood of observed data $P(data|\hat{\theta})$

$$\hat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_T + \alpha_H}$$



X=1



X=0

MLE for Bernoulli Variables

$$P(X=1) = \theta; P(X=0) = 1 - \theta$$

Data D : $\{1, 0, 0, 1, \dots\}$

Flips Produce data D with α_H heads ($X=1$) and α_T tails ($X=0$)

Flips are independent, identically distributed (which is often abbreviated to "i.i.d.").

- independent events
- identically distributed according to the Bernoulli distribution



$X=1$

$X=0$

MLE estimate: choose the value of θ that makes D most probable.

Intuition we are more likely to observe data D if we are in a world where the appearance of this data is highly probable Therefore, we should estimate θ by assigning it whatever value maximizes the probability of having observed D

MLE for Bernoulli Variables

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$$\text{Therefore } P(D|\theta) = \theta(1-\theta)(1-\theta)\theta\dots = \theta^{\alpha_H}(1-\theta)^{\alpha_T}$$

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta} P(D|\theta)$$
$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta} \ln P(D|\theta)$$



X=1



X=0

MLE for Bernoulli Variables

$$P(X=1) = \theta; P(X=0) = 1 - \theta$$

Data $D: \{1, 0, 0, 1, \dots\}$ α_H heads ($X=1$) and α_T tails ($X=0$)

$$\begin{aligned}\hat{\theta}_{MLE} &= \arg \max_{\theta} \ln P(D | \theta) \\ &= \arg \max_{\theta} \ln [\theta^{\alpha_H} (1 - \theta)^{\alpha_T}]\end{aligned}$$

Set derivative to 0. $\frac{d}{d\theta} \ln P(D | \theta) = 0$

$$\begin{aligned}\frac{d}{d\theta} \ln P(D | \theta) &= \frac{d}{d\theta} [\alpha_H \ln \theta + \alpha_T \ln(1 - \theta)] \\ &= \frac{\alpha_H}{\theta} - \frac{\alpha_T}{1 - \theta} = 0\end{aligned}$$

$$\frac{d}{d\theta} \ln \theta = \frac{1}{\theta}$$

$$\hat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_T + \alpha_H}$$



X=1



X=0

Summary: MLE for Bernoulli Variables

Problem: Assume we can flip a coin with bias θ several times. Estimate the probability that it turns out heads when we flip it?

Each flip yields a Boolean value for X , $X \sim \text{Bernoulli}(\theta)$

Bernoulli Random Variable: $P(X=1) = \theta$; $P(X=0) = 1-\theta$

Data D of independently, identically distributed (i.i.d) flips produces α_H heads ($X=1$) and α_T tails ($X=0$)

Therefore $P(D | \theta) = \theta^{\alpha_H} (1 - \theta)^{\alpha_T}$

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} P(D | \theta) = \frac{\alpha_H}{\alpha_T + \alpha_H}$$



X=1



X=0

High Probability Bound, Sample Complexity

Problem: Assume we can flip a coin with bias θ several times. Estimate the probability that it turns out heads when we flip it?

Data D: $\{1, 0, 0, 1, \dots\}$ α_H heads ($X=1$) and α_T tails ($X=0$)

$$\hat{\theta}_{MLE} = \frac{\alpha_H}{\alpha_T + \alpha_H}$$



X=1

X=0

Hoeffding Inequality: For any $\epsilon > 0$, $P\left(|\hat{\theta}_{MLE} - \theta| \geq \epsilon\right) \leq 2e^{-2n\epsilon^2}$

High Probability Bound : We want to estimate the coin parameter θ so that the probability that the estimation error exceeds ϵ is at most δ . How many coin flips are needed?

$$P\left(|\hat{\theta}_{MLE} - \theta| \geq \epsilon\right) \leq 2e^{-2n\epsilon^2} \leq \delta \quad \Rightarrow \quad n \geq \frac{\ln\left(\frac{2}{\delta}\right)}{2\epsilon^2}$$

Practice: MLE for Bernoulli Variables

A coin is flipped 10 times:

$$D = \{1, 0, 1, 1, 0, 1, 1, 1, 0, 1\}$$

where 1 = Heads and 0 = Tails.

1. Find α_H and α_T .
2. Write the likelihood function $P(D \mid \theta)$.
3. Write the log-likelihood function.
4. Find the MLE estimate $\hat{\theta}_{\text{MLE}}$.
5. Based on the MLE model, what is the estimated probability that the next flip is Heads?



X=1



X=0



Summary

- Probabilistic learning models uncertainty using $P(Y | X)$.
- A joint distribution describes probabilities over all combinations of random variables.
- Conditional probabilities can be computed from the joint distribution.
- Probability and likelihood.
- MLE chooses the parameter that makes the observed data most likely.